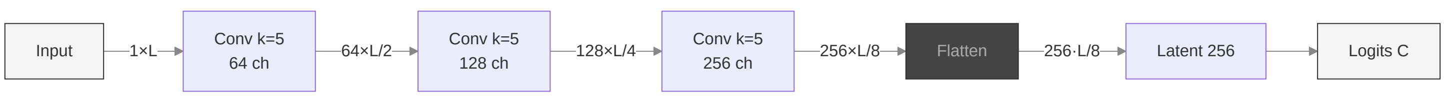
Model Zoo — Architecture Diagrams

All 10 models take a 1D signal `(batch, 1, L)` and output class logits `(batch, C)`.

Legend. Edge labels show the tensor shape `channels×length` at each step. Signature blocks are highlighted in the family colour.

1. Conv1D

Plain CNN — 3 conv blocks, flatten, FC head.



> Block = Conv1d + BN + ReLU + MaxPool(/2) + Dropout. **Flatten** couples head size to L.

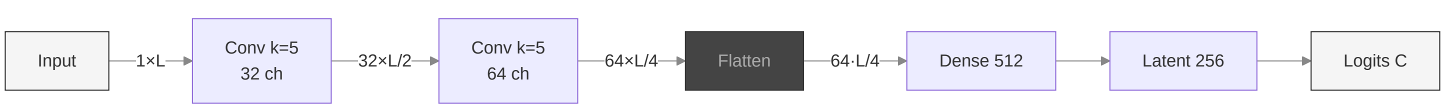
2. Conv1DGAP

Same backbone as Conv1D but **Global Average Pooling** replaces flatten → input-length agnostic, much smaller head.

> **GAP** = AdaptiveAvgPool1d(1) — collapses the temporal axis to 1, regardless of L.

3. LeNet1D

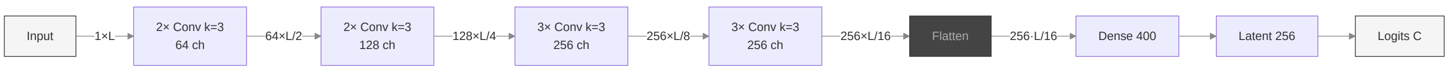
Classic LeNet-5 adapted to 1D — 2 conv stages, heavy FC head.



> Minimal backbone, heavy FC head — parameter count scales with L.

4. VGG1D

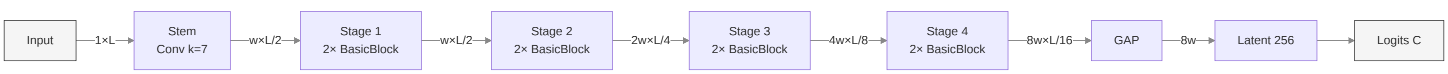
Stacked small (k=3) convolutions, doubling channels — 4 blocks.

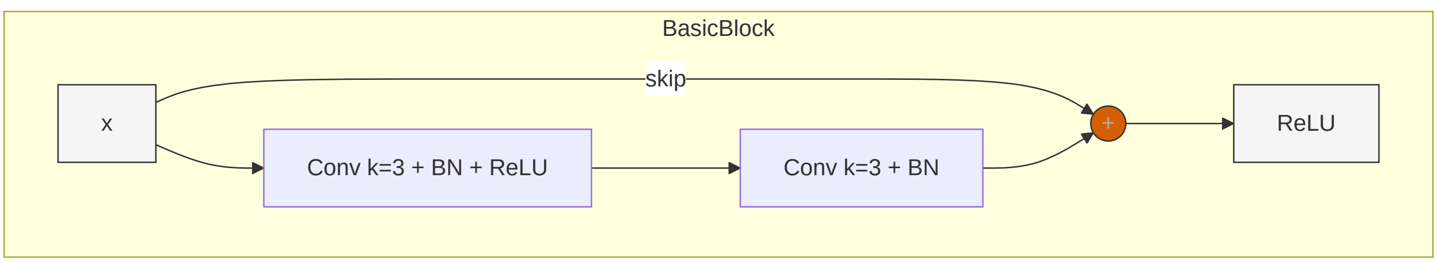


> Block = N × (Conv + BN + ReLU) + MaxPool(/2). Heavy flatten → large FC.

5. ResNet1D

Residual blocks with **skip connections** — ResNet-18 style. Width `w = base_width` (default 74).

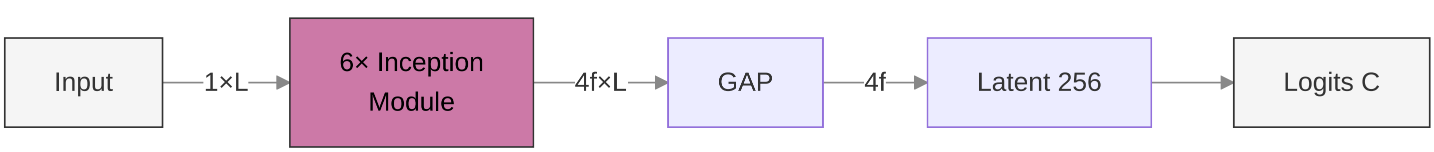


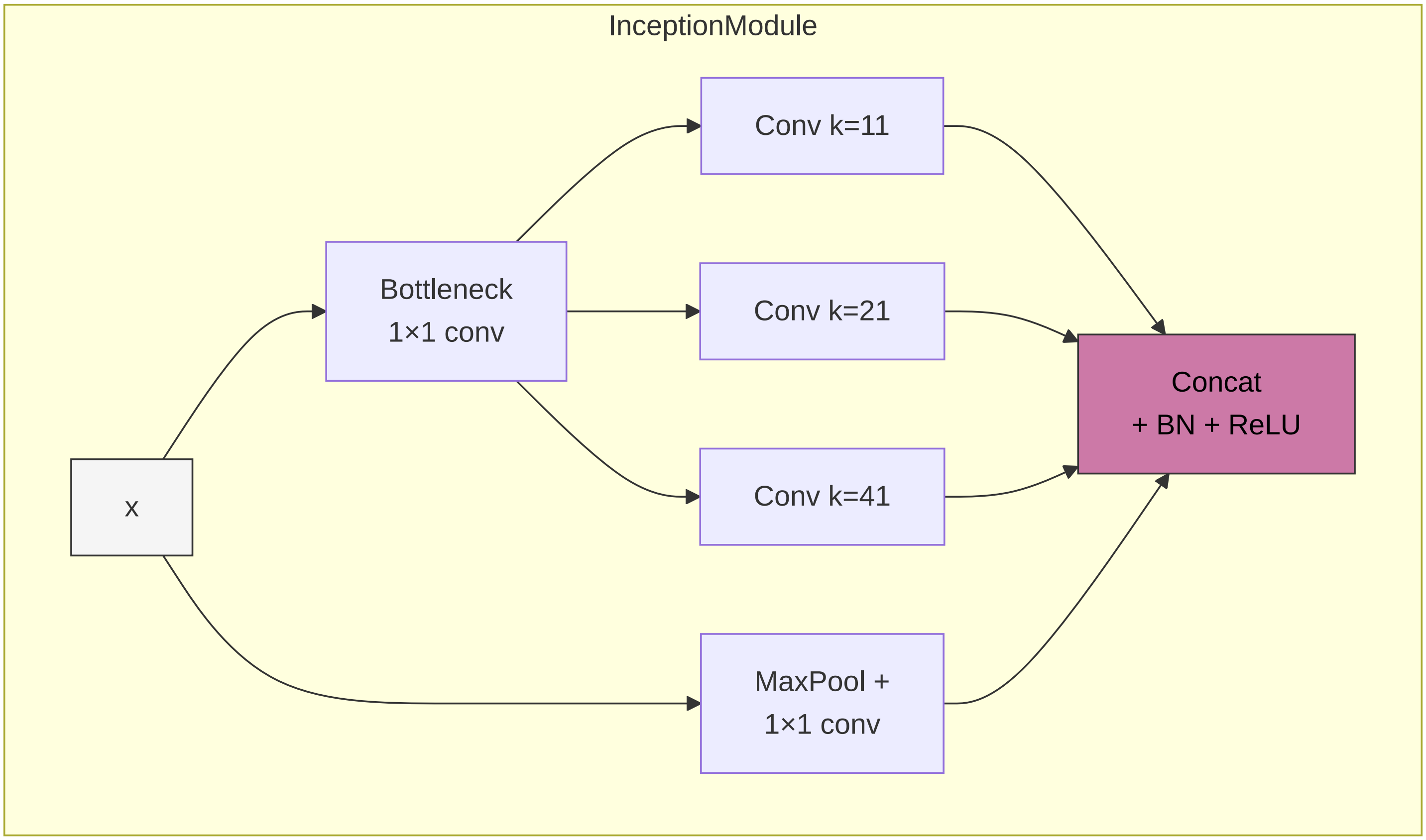


> Stage 2+ doubles channels and halves length via strided first block. Identity skip when shapes match, else 1×1 projection.

6. InceptionTime1D

Multi-scale parallel convolutions — 6 modules, residual shortcut every 3.

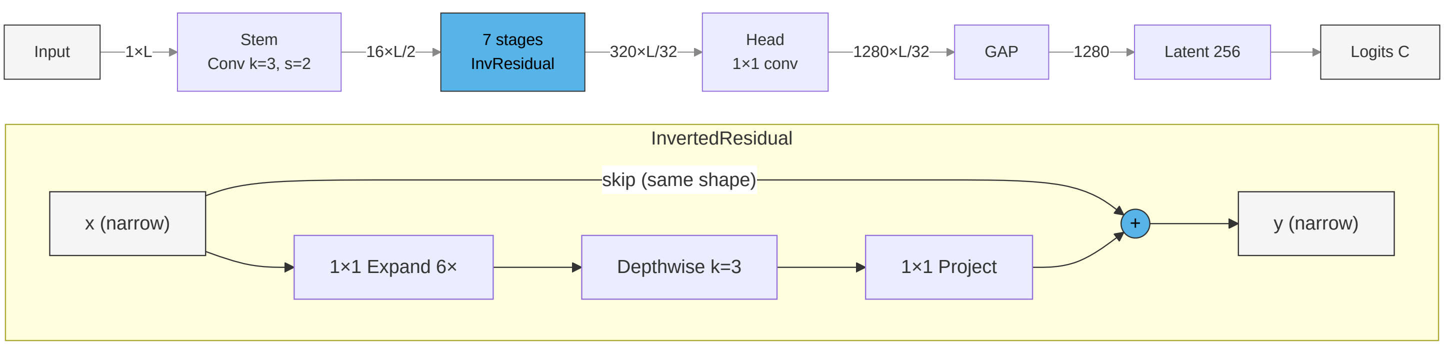




> Length preserved inside the module; output = concat of 4 parallel branches (→ 4f channels). `f = num_filters` (default 148).

7. MobileNet1D

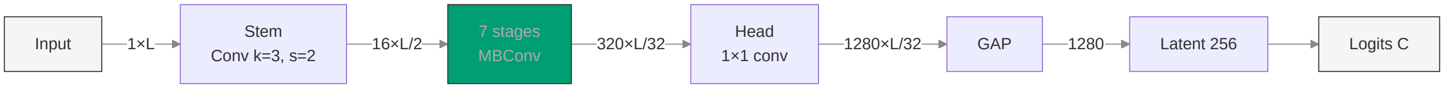
Depthwise separable convolutions with inverted residual bottleneck (MobileNetV2).

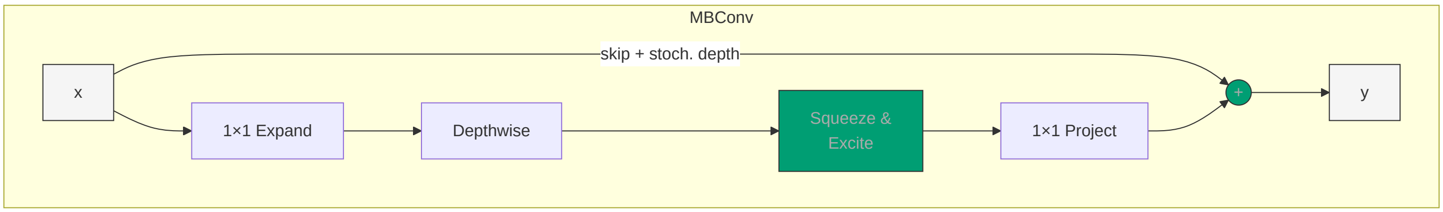


> ReLU6 activations. Linear bottleneck (no activation after project). Skip only when input/output shapes match.

8. EfficientNet1D

Like MobileNet but adds **Squeeze-and-Excitation** + **stochastic depth** (EfficientNet-B0 style).

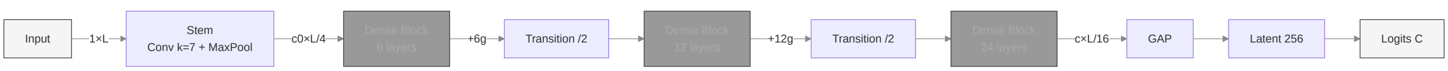


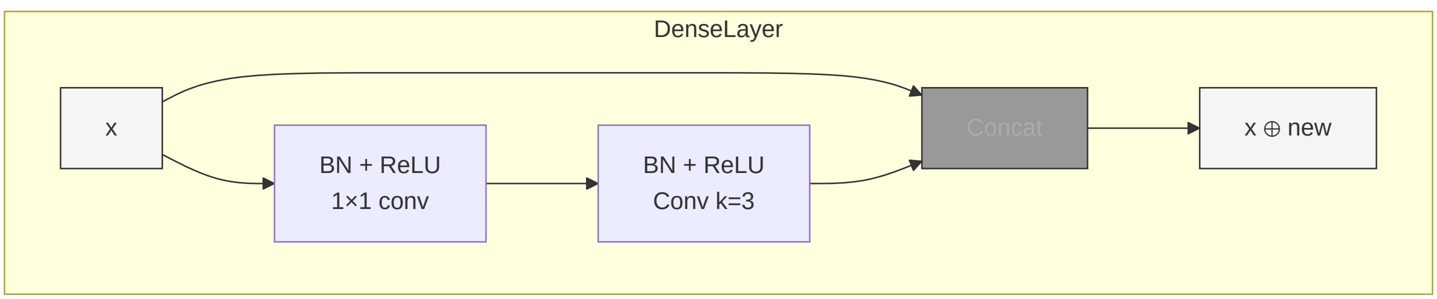


> SiLU activations. SE recalibrates channel importance. Stochastic depth drops entire blocks during training.

9. DenseNet1D

Dense connectivity — each layer receives all preceding feature maps via concatenation.

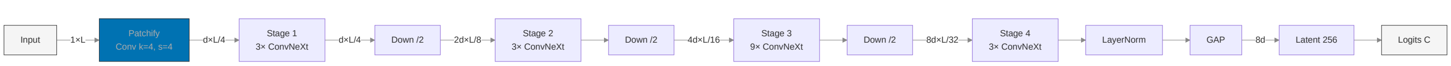


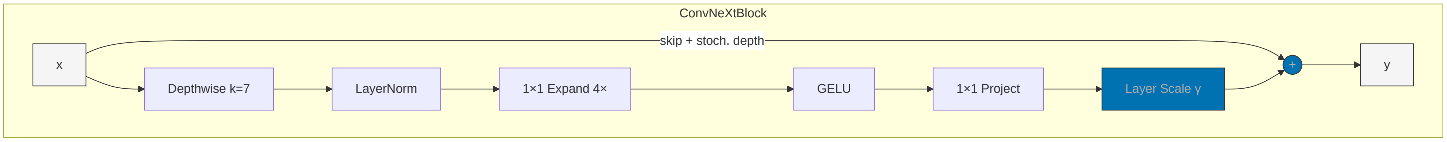


> Growth rate `g` = new channels per layer. Transition = BN + 1×1 conv + AvgPool (halves channels + length).

10. ConvNeXt1D

Modernised CNN inspired by Transformers — patchify stem, large-kernel DWConv, inverted bottleneck MLP, LayerNorm + GELU, layer scale.





> Depths [3, 3, 9, 3] (ConvNeXt-Tiny). Channels double per stage: [d, 2d, 4d, 8d], with d = `base_dim` (default 42). Per-channel layer scale γ (init 1e-6) stabilises training.

Quick Comparison

Model	Key Idea	Conv backbone	Classifier head	Head params (→ Latent 256)
Conv1D	Plain CNN	3 × Conv k=5, 64→256 ch	Flatten(256·L/8) → 256 → C	8192·L
Conv1DGAP	Plain CNN + GAP	3 × Conv k=5, 64→256 ch	GAP(256) → 256 → C	66k
LeNet1D	Minimal (LeNet-5)	2 × Conv k=5, 32→64 ch	Flatten(64·L/4) → 512 → 256 → C	8192·L + 132k
VGG1D	Stacked k=3	10 Conv k=3 (4 blocks), 64→256 ch	Flatten(256·L/16) → 400 → 256 → C	6400·L + 103k
ResNet1D	Skip connections	8 blocks × k=3, w→8w ch	GAP(8w) → 256 → C	2048·w
InceptionTime1D	Multi-scale parallel	6 modules × (k=11,21,41), 4f ch	GAP(4f) → 256 → C	1024·f
MobileNet1D	Depthwise separable	17 InvRes blocks, 16→320 ch	GAP(1280) → 256 → C	328k
EfficientNet1D	MBConv + SE	16 MBConv blocks, 16→320 ch	GAP(1280) → 256 → C	328k

Model	Key Idea	Conv backbone	Classifier head	Head params (→ Latent 256)
DenseNet1D	Dense concat	42 layers (3 blocks), +g ch/layer	GAP(ch) → 256 → C	256·ch_final
ConvNeXt1D	Modernised CNN (DWConv k=7 + MLP)	18 blocks (4 stages [3,3,9,3]), d→8d ch	GAP(8d) → 256 → C	2048·d

> **Flatten-based heads** (Conv1D, LeNet, VGG) have parameter counts that scale with input length L, while **GAP-based heads** are L-agnostic.